ECO-SYNC

Deforestation & Illegal Logging Early Detection System

SDG 15: LIFE ON LAND

TEAM ID	EVENT	INSTITUTION	ORGANIZATION
TFT002	CMR ThinkFest 3.0 (Aug 2026)	CMR College of Eng. & Tech.	Centre for CEER

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INTELLIGENT ACOUSTIC MONITORING	CELLULAR / INTERNET NEEDED AT EDGE	HARDWARE VS SOFTWARE REVENUE RATIO

1. EXECUTIVE SUMMARY

Eco-Sync turns remote, endangered forest canopies into an intelligent 24/7 listening network capable of stopping illegal deforestation before trees hit the ground. Operating entirely on local Edge AI, tree-mounted listening nodes analyze ambient acoustics to detect threat signatures such as chainsaws and gunshots within seconds. By combining ultra-low-power embedded hardware, long-range LoRa radio communication, and automated SMS GPS routing, Eco-Sync provides underfunded forest guardians with real-time actionable intelligence—eliminating cellular coverage blind spots and high foot-patrol overhead.

2. PROBLEM STATEMENT & TECHNICAL SOLUTION

The Core Problem

Silent, rapid forest destruction plagues critical biodiversity hotspots worldwide due to severe structural limitations:

- **Zero Coverage Blind Spots:** Deep reserves lack cellular or internet connectivity, leaving rangers blind.
- **Delayed Threat Intelligence:** Satellite imagery provides reporting delays of days or weeks.
- **Astronomical Patrol Costs:** Manual, blind foot patrols exhaust underfunded forestry departments without preventing logging in real time.

The Eco-Sync Solution

An end-to-end off-grid acoustic intelligence ecosystem:

- **Edge AI Acoustic Processing:** Local microcontrollers analyze canopy acoustics natively without transmitting raw audio.
- **LoRa Long-Range Mesh:** Verified threat alerts are sent via long-range LoRa radio to a central gateway.
- **Instant GPS Routing:** Central gateway converts alerts into SMS notifications containing exact GPS coordinates straight to ranger mobile phones.

System Architecture Blueprint

[Tree Canopy Node] (Edge AI filtering & acoustic analysis) → **[LoRa Transmission]** (Off-grid long range) → **[Central Gateway]** → **[Instant SMS Alert]** (Exact GPS coordinates directly to ranger handset).

3. BUSINESS MODEL CANVAS ANALYSIS

The Eco-Sync business framework is designed to bridge cutting-edge technology with the operational realities of underfunded environmental protection agencies.

Building Block	Strategic Implementation Details
1. Customer Segments	**Primary:** Government forestry departments and national park authorities protecting vast reserves. **Secondary:** Global conservation NGOs (e.g., WWF, Rainforest Alliance) and frontline indigenous community forest guardians.
2. Value Proposition	Provides an instant, zero-blindspot tactical advantage in areas without cellular coverage. Transforms delayed disaster management into proactive, real-time interception, saving thousands of forest acres on a fraction of traditional patrol budgets.
3. Channels	Direct partnerships with regional conservation networks, on-site pilot deployments, field demonstration kits, and international hardware distribution channels.
4. Customer Relationships	High-touch, collaborative co-creation involving dedicated technical support and hands-on ranger training. Shifts over time to low-cost automated remote maintenance and portal management.
5. Revenue Streams	**60% - Hardware Sales:** Weatherproof canopy nodes, signal receivers, and deployment kits funded via upfront CAPEX/grants. **40% - Software & Support:** Annual recurring subscriptions for AI software updates, maintenance, and platform licensing.
6. Key Resources	Custom Edge AI acoustic algorithms, ruggedized outdoor hardware, specialized embedded-systems engineers, field support teams, and conservation logistics networks.
7. Key Activities	Manufacturing weatherproof canopy nodes, continuous ML acoustic model training, ultra-low power hardware optimization, and ranger onboarding field logistics.
8. Key Partners	Government forestry agencies, international NGOs (WWF, Rainforest Alliance), indigenous land management groups, microchip vendors, and solar/battery component manufacturers.
9. Cost Structure	Hardware assembly/manufacturing, embedded hardware components (microchips, sensors, solar panels), acoustic AI model fine-tuning R&D, and field deployment/ranger training logistics.

4. FINANCIAL DYNAMICS & STRATEGIC SUSTAINABILITY

Revenue Stream Split

Hardware Sales (60%): Purchased via annual government capital budgets, multilateral conservation grants, and donor allocations. Includes node hardware, LoRa gateways, and mounting gear.

SaaS & Maintenance (40%): Predictable annual subscription fees covering remote software optimization, model update deployments, and ongoing technical support.

Cost & Risk Drivers

Primary Cost Drivers: R&D talent acquisition in embedded systems/AI, custom weatherproof node enclosure manufacturing, and initial field onboarding logistics in remote geographies.

Risk Mitigation: Shifting high-touch onboarding to standardized self-service ranger training toolkits over time reduces field overhead.

5. ENVIRONMENTAL IMPACT & UN SDG 15 ALIGNMENT

Eco-Sync directly advances **United Nations Sustainable Development Goal 15 (Life on Land)** by targeting target 15.2 (halt deforestation) and target 15.7 (combat poaching and illegal trafficking).

- **Proactive Conservation:** Replaces post-facto satellite analysis with real-time intervention before trees are felled.
- **Empowerment of Local Communities:** Gives indigenous forest stewards real-time actionable intelligence, leveling the field against illegal logging syndicates.
- **Biodiversity Preservation:** Shields critical ecosystem habitats from noise pollution, habitat fragmentation, and poaching threats.

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